

Getting Started with pgscatalogcolors

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```
library(pgscatalogcolors)
library(ggplot2)
```

What ancestry acronyms are available?

The package maps **19 ancestry categories** defined by the PGS Catalog ancestry documentation. Call `pgscatalog_ancestry_info()` to see the full lookup table:

```
pgscatalog_ancestry_info()
#>   acronym          ancestry pop_group  color
#> 1   ABO             Aboriginal Australian  DIV #999999
#> 2   AAM African American or Afro-Caribbean  AFR #F9DA49
#> 3   AFR             African unspecified  AFR #F9DA49
#> 4   ASN             Asian unspecified  ASN #B25928
#> 5   CAS             Central Asian  ASN #B25928
#> 6   EAS             East Asian  EAS #68AD57
#> 7   EUR             European  EUR #377EB8
#> 8   MID             Greater Middle Eastern  MID #5DCBCF
#> 9   HIS             Hispanic or Latin American  HIS #D1352B
#> 10  AMR             Native American  DIV #999999
#> 11  NON             Not Reported  NON #BBBBBB
#> 12  OCE             Oceanian  DIV #999999
#> 13  OTH             Other  DIV #999999
#> 14  ADA             Other admixed ancestry  DIV #999999
#> 15  SAS             South Asian  SAS #984EA4
#> 16  SEA             South East Asian  ASN #B25928
#> 17  SAF             Sub-Saharan African  AFR #F9DA49
#> 18  MIE Multi-Ancestry including Europeans  MIE #E887BD
#> 19  MEE Multi-Ancestry excluding Europeans  MEE #EF8632
```

The columns are:

Column	Description
acronym	The exact string your data column must contain
ancestry	Full PGS Catalog ancestry label
pop_group	Broader population group abbreviation (1000G/HGDP-style)
color	Hex color assigned to that acronym

Column	Description
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What does my data frame need?

Your data frame needs **one column whose values are ancestry acronyms** — the exact strings from the `acronym` column above (case-sensitive). The column can have any name; you point `ggplot2` to it with `aes(color = <your_column>)`.

```
# Minimal working example
df <- data.frame(
  sample_id = paste0("s", 1:6),
  pgs_score = c(0.42, -0.11, 0.87, 0.03, -0.55, 0.29),
  ancestry = c("EUR", "AFR", "EAS", "SAS", "HIS", "MID")
  #           ^-- must match acronyms exactly
)
df
#>   sample_id pgs_score ancestry
#> 1         s1      0.42      EUR
#> 2         s2     -0.11      AFR
#> 3         s3      0.87      EAS
#> 4         s4      0.03      SAS
#> 5         s5     -0.55      HIS
#> 6         s6      0.29      MID
```

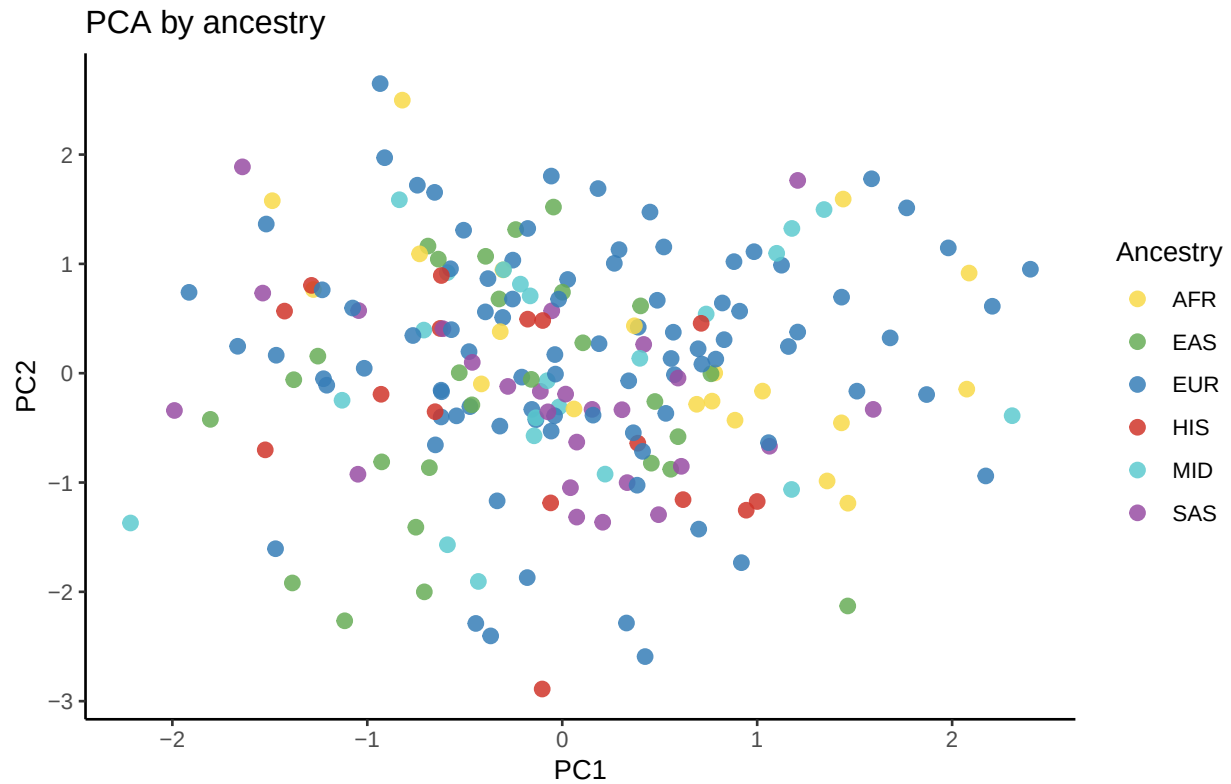
The `ancestry` column values must be **character or factor** — numeric codes will not match. The column does **not** need to contain all 19 acronyms; only the ones present in your data are plotted.

Basic scatter plot

```
set.seed(1)
n <- 200
grp <- c("EUR", "AFR", "EAS", "SAS", "HIS", "MID")

df <- data.frame(
  PC1 = rnorm(n),
  PC2 = rnorm(n),
  ancestry = sample(grp, n, replace = TRUE,
                    prob = c(0.40, 0.15, 0.15, 0.12, 0.10, 0.08))
)

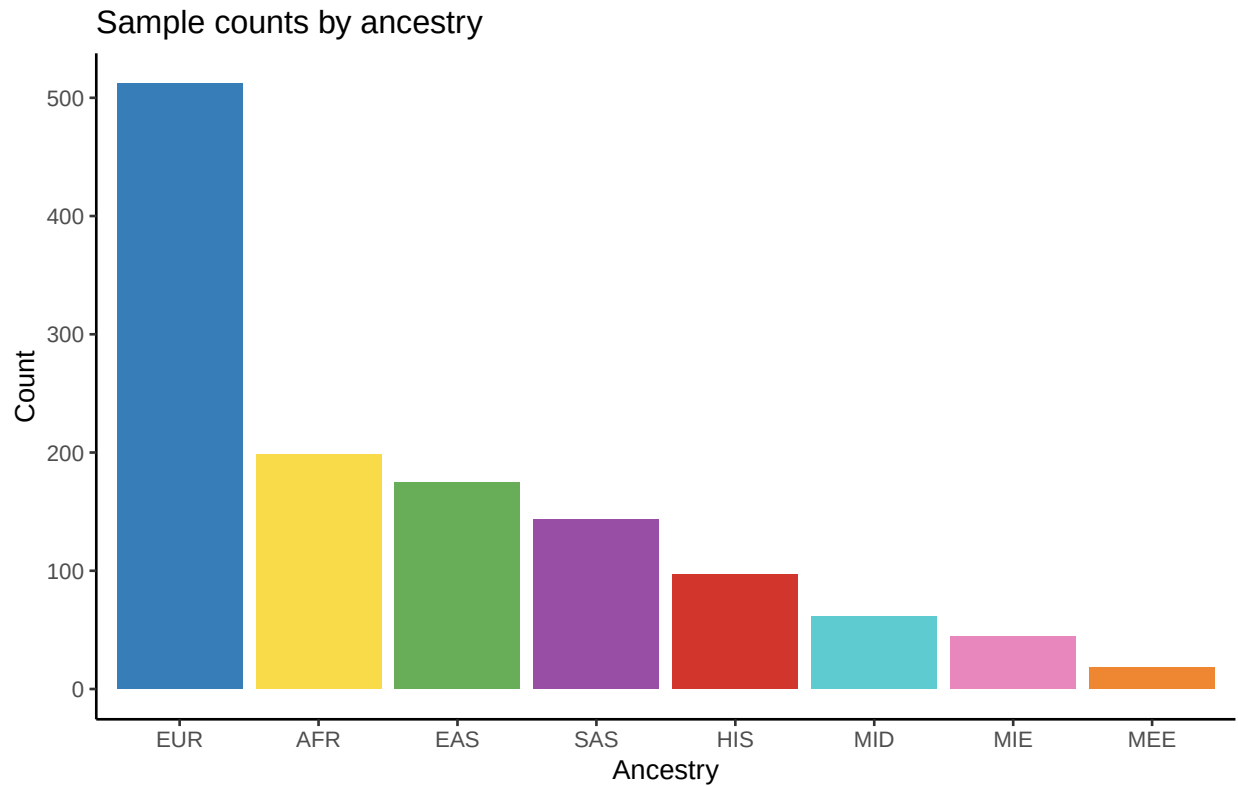
ggplot(df, aes(PC1, PC2, color = ancestry)) +
  geom_point(size = 2.5, alpha = 0.85) +
  scale_color_pgs catalog() +
  labs(title = "PCA by ancestry", color = "Ancestry") +
  theme_classic()
```



Bar chart (fill aesthetic)

```
counts <- data.frame(
  ancestry = c("EUR", "AFR", "EAS", "SAS", "HIS", "MID", "MIE", "MEE"),
  n        = c(512, 198, 175, 143, 97, 61, 44, 18)
)

ggplot(counts, aes(reorder(ancestry, -n), n, fill = ancestry)) +
  geom_col() +
  scale_fill_pgscatalog() +
  labs(x = "Ancestry", y = "Count", title = "Sample counts by ancestry") +
  theme_classic() +
  theme(legend.position = "none")
```

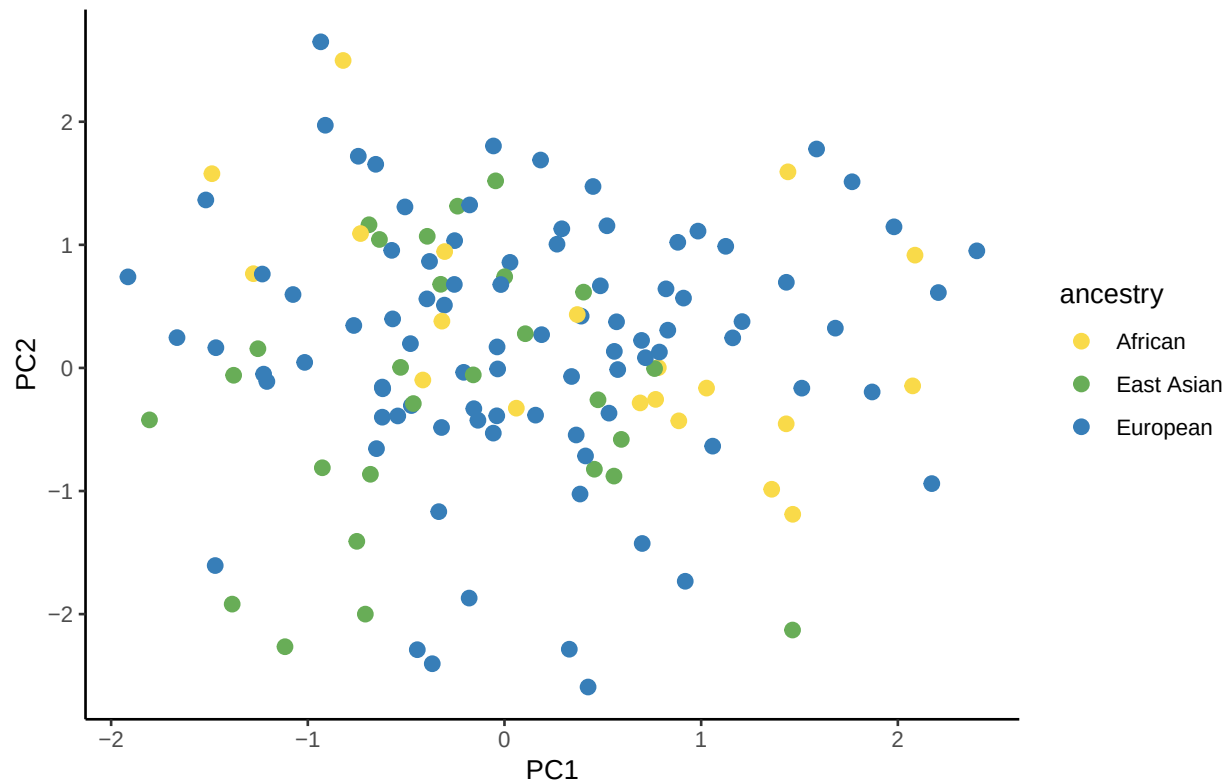


Using `scale_color_manual()` directly

If you need more control (e.g., custom legend labels), use `pgscatalog_scale_values()` to get only the colors for groups in your data:

```
my_groups <- c("EUR", "AFR", "EAS")

ggplot(df[df$ancestry %in% my_groups, ], aes(PC1, PC2, color = ancestry)) +
  geom_point(size = 2.5) +
  scale_color_manual(
    values = pgscatalog_scale_values(my_groups),
    labels = c(AFR = "African", EAS = "East Asian", EUR = "European")
  ) +
  theme_classic()
```



Checking colors programmatically

```
# Look up specific colors by acronym
pgscatalog_cols("EUR", "AFR", "EAS")
#>      EUR      AFR      EAS
#> "#377EB8" "#F9DA49" "#68AD57"

# Check whether your acronyms are all recognized before plotting
my_data_groups <- c("EUR", "AFR", "XYZ") # "XYZ" is not a valid acronym
pgscatalog_scale_values(my_data_groups) # warns about unknown acronyms
#> Warning in pgscatalog_scale_values(my_data_groups): Unknown acronyms (will use
#> grey): XYZ
#>      EUR      AFR      XYZ
#> "#377EB8" "#F9DA49" "#999999"
```

What if my data uses full ancestry names instead of acronyms?

Join against `pgscatalog_ancestry_info()` to add the acronym column:

```
raw <- data.frame(
  sample_id = 1:3,
  ancestry_long = c("European", "African unspecified", "East Asian")
)

info <- pgscatalog_ancestry_info()
merged <- merge(raw, info[, c("ancestry", "acronym")],
  by.x = "ancestry_long", by.y = "ancestry")
merged
```

```
#>      ancestry_long sample_id acronym
#> 1 African unspecified      2    AFR
#> 2      East Asian      3    EAS
#> 3      European      1    EUR
```